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- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Yanzhu Guo, Guokan Shang, and Chloé Clavel. Benchmarking linguistic diversity of large language models. *arXiv preprint arXiv:2412.10271*, 2024.
- Neha Gupta, Harikrishna Narasimhan, Wittawat Jitkittum, Ankit Singh Rawat, Aditya Krishna Menon, and Sanjiv Kumar. Language model cascades: Token-level uncertainty and beyond. In *The Twelfth International Conference on Learning Representations*, 2024.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Hong, Zhen Wang, Daisy Wang, and Zhiting Hu. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 8154–8173, 2023.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=d7KBjmI3GmQ>.
- John Hewitt, Christopher D Manning, and Percy Liang. Truncation sampling as language model desmoothing. In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pp. 3414–3427, 2022.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=rygGQyrFvH>.
- Mete Ismayilzada, Antonio Laverghetta Jr, Simone A Luchini, Reet Patel, Antoine Bosselut, Lonneke van der Plas, and Roger Beaty. Creative preference optimization. *arXiv preprint arXiv:2505.14442*, 2025.
- T Jaeger and Roger Levy. Speakers optimize information density through syntactic reduction. *Advances in neural information processing systems*, 19, 2006.
- Zhengbao Jiang, Jun Araki, Haibo Ding, and Graham Neubig. How can we know when language models know? on the calibration of language models for question answering. *Transactions of the Association for Computational Linguistics*, 9:962–977, 2021.
- Daniel Jurafsky and James H. Martin. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, with Language Models*. 3rd edition, 2025. URL <https://web.stanford.edu/~jurafsky/slp3/>. Online manuscript released August 24, 2025.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=PX3FAVHJT>.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *The Eleventh International Conference on Learning Representations*, 2023.
- Oh Joon Kwon, Daiki E Matsunaga, and Kee-Eung Kim. Gdpo: Learning to directly align language models with diversity using gflownets. *arXiv preprint arXiv:2410.15096*, 2024.
- Thom Lake, Eunsol Choi, and Greg Durrett. From distributional to overton pluralism: Investigating large language model alignment. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 6794–6814, 2025.

- 
- Jack Lanchantin, Angelica Chen, Shehzaad Dhuliawala, Ping Yu, Jason Weston, Sainbayar Sukhbaatar, and Ilia Kulikov. Diverse preference optimization. *arXiv preprint arXiv:2501.18101*, 2025.
- Roger Levy. Expectation-based syntactic comprehension. *Cognition*, 106(3):1126–1177, 2008.
- Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and William B Dolan. A diversity-promoting objective function for neural conversation models. In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 110–119, 2016.
- Margaret Li, Weijia Shi, Artidoro Pagnoni, Peter West, and Ari Holtzman. Predicting vs. acting: A trade-off between world modeling & agent modeling. *arXiv preprint arXiv:2407.02446*, 2024a.
- Yucheng Li, Frank Guerin, and Chenghua Lin. Latesteval: Addressing data contamination in language model evaluation through dynamic and time-sensitive test construction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 18600–18607, 2024b.
- Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base llms: Rethinking alignment via in-context learning. In *The Twelfth International Conference on Learning Representations*, 2023.
- Siyang Liu, Sahand Sabour, Yinhe Zheng, Pei Ke, Xiaoyan Zhu, and Minlie Huang. Rethinking and refining the distinct metric. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 762–770, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-short.86. URL <https://aclanthology.org/2022.acl-short.86/>.
- Li-Chun Lu, Shou-Jen Chen, Tsung-Min Pai, Chan-Hung Yu, Hung-yi Lee, and Shao-Hua Sun. Llm discussion: Enhancing the creativity of large language models via discussion framework and role-play. In *First Conference on Language Modeling*, 2024.
- Ximing Lu, Melanie Sclar, Skyler Hallinan, Niloofer Mireshghallah, Jiacheng Liu, Seungju Han, Allyson Ettinger, Liwei Jiang, Khyathi Chandu, Nouha Dziri, and Yejin Choi. AI as humanity’s salieri: Quantifying linguistic creativity of language models via systematic attribution of machine text against web text. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=i10EOIqolQ>.
- Kyle Mahowald, Evelina Fedorenko, Steven T Piantadosi, and Edward Gibson. Info/information theory: Speakers choose shorter words in predictive contexts. *Cognition*, 126(2):313–318, 2013.
- Clara Meister, Tiago Pimentel, Patrick Haller, Lena Jäger, Ryan Cotterell, and Roger Levy. Revisiting the uniform information density hypothesis. *arXiv preprint arXiv:2109.11635*, 2021.
- William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023.
- George A Miller. Wordnet: a lexical database for english. *Communications of the ACM*, 38(11):39–41, 1995.
- Avinash Mudireddy, Tyler Bell, and Raghu Mudumbai. Slaves to the law of large numbers: An asymptotic equipartition property for perplexity in generative language models. *arXiv preprint arXiv:2405.13798*, 2024.
- Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. One fish, two fish, but not the whole sea: Alignment reduces language models’ conceptual diversity. In Luis Chiruzzo, Alan Ritter, and Lu Wang (eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 11241–11258, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.561. URL <https://aclanthology.org/2025.naacl-long.561/>.

- 
- Shashi Narayan, Shay B Cohen, and Mirella Lapata. Don't give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pp. 1797–1807, 2018.
- Brendan O'Connor. Perplexity as branching factor as shannon diversity index, 2013. URL <https://brenocon.com/blog/2013/01/perplexity-as-branching-factor-as-shannon-diversity-index/>. Accessed: 2024-09-28.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, et al. 2 olmo 2 furious. *arXiv preprint arXiv:2501.00656*, 2024.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Vishakh Padmakumar and He He. Does writing with language models reduce content diversity? In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Feiz5HtCD0>.
- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, pp. 548–560, 2023.
- Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. Red teaming language models with language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 3419–3448, 2022.
- Team Qwen. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.
- Matthew Renze and Erhan Guven. The effect of sampling temperature on problem solving in large language models. *arXiv preprint arXiv:2402.05201*, 2024.
- Julian Rodemann, Esteban Garces Arias, Christoph Luther, Christoph Jansen, and Thomas Augustin. A statistical case against empirical human-ai alignment. *arXiv preprint arXiv:2502.14581*, 2025.
- Abulhair Saparov and He He. Language models are greedy reasoners: A systematic formal analysis of chain-of-thought. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=qFVVBzXxR2V>.
- Claude E Shannon. A mathematical theory of communication. *The Bell system technical journal*, 27(3): 379–423, 1948.
- Chufan Shi, Haoran Yang, Deng Cai, Zhisong Zhang, Yifan Wang, Yujiu Yang, and Wai Lam. A thorough examination of decoding methods in the era of llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 8601–8629, 2024a.
- Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. Detecting pretraining data from large language models. In *The Twelfth International Conference on Learning Representations*, 2024b.
- Shivalika Singh, Freddie Vargas, Daniel D'souza, Börje Karlsson, Abinaya Mahendiran, Wei-Yin Ko, Herumb Shandilya, Jay Patel, Deividas Mataciunas, Laura O'Mahony, et al. Aya dataset: An open-access collection for multilingual instruction tuning. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11521–11567, 2024.